

Assignment

Note: Solve all the questions and write your answers clearly, neatly, and correctly on the paper.

1. A hospital recorded the waiting time (in minutes) of patients in the emergency department as follows:
Waiting Time (minutes) 0–10 10–20 20–30 30–40 40–50
Number of Patients 12 25 18 10 5
Find the average (mean) waiting time per patient.
2. A university conducted a student feedback survey on teaching quality. The satisfaction scores (out of 100) are grouped as follows:
Score Range. 0–25 25–50 50–75 75–100
Number of Students 8 22 45 15
Determine the median satisfaction score.
3. The runs scored by two batsmen A and B in 10 cricket matches are given below:
Batsman A: 48, 65, 32, 55, 70, 40, 52, 60, 38, 45, Batsman B: 25, 80, 35, 15, 90, 55, 42, 20, 85, 30. Calculate the standard deviation for both batsmen and determine who is more consistent using the coefficient of variation.
4. A college has 18 students from the Science stream and 7 students from the Commerce stream. Five students are selected at random without replacement to represent the college at a national youth summit. Find the probability that: (a) All five selected students are from the Science stream. (b) The first three selected are from Science and the last two are from Commerce.
5. In a city, 65% of the residents have health insurance. Among the insured residents, 3% suffer from a particular chronic condition, while 12% of the uninsured residents suffer from the same condition. If a randomly selected resident is found to have this chronic condition, use Bayes' Theorem to find the probability that the person has health insurance.
6. In a competitive exam, the probability that a candidate answers a multiple-choice question correctly is 0.75. A candidate attempts 10 questions independently. Find: (a) The probability that exactly 7 questions are answered correctly. (b) The probability that at least 8 questions are answered correctly.
7. The weights of newborn babies in a hospital are approximately normally distributed with a mean of 3.2 kg and a standard deviation of 0.4 kg. (a) What percentage of babies weigh less than 2.6 kg? (b) What percentage of babies weigh between 2.8 kg and 3.6 kg? (c) What percentage of babies weigh more than 3.8 kg?
8. A researcher wants to study the relationship between daily study hours and exam scores among 12 students. The data is given below:

Study Hours (X)	4	6	3	8	5	7	2	9	4	6	5	7
Exam Score (Y)	55	72	48	85	65	78	40	90	58	70	62	82

Calculate Spearman's Rank Correlation Coefficient and interpret the relationship.

9. A car manufacturer wants to study the relationship between the speed of a vehicle and fuel consumption during a test drive.

Speed in km/h (X)	30	40	50	60	70	80	90	100	110	120
Fuel Consumption in litres (Y)	3	4	5	6	7	8	9	10	11	12

Calculate Karl Pearson's coefficient of correlation between vehicle speed and fuel consumption, and interpret the relationship between the two variables.

10. The following data shows the advertising expenditure (in ₹ lakhs) and sales (in ₹ crores) of a company over 10 months:

Adv. Expenditure (X) 12 15 18 10 22 25 14 20 16 19

Sales (Y) 45 52 68 38 75 88 50 70 55 62

- (a) Find the regression equation of sales (Y) on advertising expenditure (X).
(b) Estimate the expected sales when the advertising expenditure is ₹23 lakhs.

11. A fitness company introduced a new diet plan and wants to test whether it produces greater weight loss than the old plan. A sample of 50 participants on the new plan lost an average of 5.6 kg with a standard deviation of 1.4 kg. A sample of 48 participants on the old plan lost an average of 4.7 kg with a standard deviation of 1.6 kg. Test at the 5% significance level whether the new diet plan results in significantly greater weight loss.

12. A smartphone company claims that at least 75% of its customers are satisfied with the after-sales service. In a random sample of 250 customers, 170 reported being satisfied. Test at the 5% significance level whether there is sufficient evidence to reject the company's claim (i.e., whether the true satisfaction proportion is less than 75%).

13. A fitness coach wants to compare the effectiveness of two diet plans on weight reduction. A sample of 45 people following Diet Plan A lost an average of 6.8 kg with a standard deviation of 1.9 kg, while 40 people following Diet Plan B lost an average of 5.9 kg with a standard deviation of 2.1 kg. Test at the 5% level of significance whether Diet Plan A leads to greater weight reduction than Diet Plan B.

14. A mobile phone company claims that the average battery life of its smartphones is 20 hours. A random sample of 14 phones shows an average battery life of 18.2 hours with a standard deviation of 2.5 hours. At the 1% significance level, test whether the average battery life is less than 20 hours. Assume the population is normally distributed.

15. A university conducted a survey to study the relationship between students' academic department and their satisfaction with online learning facilities. The data collected are shown below:

Satisfaction Level	Science	Commerce	Arts	Engineering
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Satisfied	42	36	28	40
Not Satisfied	18	24	22	20

At the 5% significance level, test whether academic department and satisfaction with online learning facilities are associated.

16. Two training programs are compared using employee productivity scores. Fifteen employees trained under Program A obtained an average productivity score of 82 with a standard deviation of 7, while 13 employees trained under Program B obtained an average score of 76 with a standard deviation of 6. At the 5% level of significance, test whether the mean productivity score of Program A exceeds that of Program B by more than 4 points. Assume normal populations with equal variances.
17. A supermarket records the number of defective items found in groups of 5 products. The following data were collected from 180 groups:

Number of Defects	0	1	2	3	4	5
Frequency	42.	68	44	18	6	2

Using the chi-square goodness-of-fit test at the 5% significance level, determine whether the data follow a Binomial distribution.

18. Three fertilizer types are tested to determine their effect on crop yield in an agricultural field. The increase in crop yield (in quintals per acre) is recorded below:
- Fertilizer A 14 16 18 15 17
- Fertilizer B 8 10 9 7 11
- Fertilizer C 12 13 15 14 16

Using one-way ANOVA at the 5% significance level, test whether the mean crop yield differs among the three fertilizer types.

19. A manufacturing company wants to compare the consistency of production times from two assembly lines. A sample of 18 observations from Assembly Line A has a standard deviation of 5.8 minutes, while a sample of 15 observations from Assembly Line B has a standard deviation of 4.6 minutes. At the 5% significance level, test whether the two assembly lines have equal variability in production time.